

lab01

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First part

1. Run *nslookup* to obtain the IP address of a Web server in Asia.

Ans.

```
[pin@MacBook-khxng-Nathan ~ % nslookup www.chula.ac.th
Server:          192.168.64.3
Address:         192.168.64.3#53

Non-authoritative answer:
www.chula.ac.th canonical name = 96owsv2.x.incapdns.net.
Name:   96owsv2.x.incapdns.net
Address: 45.60.126.77

pin@MacBook-khxng-Nathan ~ %
```

DNS (Domain Name System) is like a phone book, a name of a location, but IP is like a phone number

1. Non-authoritative answer (this is the information it gets from cache of a local DNS server (most of the time, it is a local router))

2. Run *nslookup* to determine the authoritative DNS servers for a university in Europe.

Ans.

Authoritative-answers (answer from local DNS server, basically a local phonebook) **name server**

Authoritative name, name from **actual** server, and the authoritative server does not hold cache as well.

With the request (nslookup) for authoritative server, we are asking for information (IP

address) from actual server itself, not cache

```
[pin@MacBook-khxng-Nathan ~ % nslookup -type=NS ox.ac.uk
Server:          192.168.64.3
Address:         192.168.64.3#53

Non-authoritative answer:
ox.ac.uk         nameserver = dns2.ox.ac.uk.
ox.ac.uk         nameserver = dns0.ox.ac.uk.
ox.ac.uk         nameserver = dns1.ox.ac.uk.
ox.ac.uk         nameserver = auth6.dns.ox.ac.uk.
ox.ac.uk         nameserver = auth5.dns.ox.ac.uk.
ox.ac.uk         nameserver = auth4.dns.ox.ac.uk.

Authoritative answers can be found from:

pin@MacBook-khxng-Nathan ~ %
```

3. Run *nslookup* so that one of the DNS servers obtained in Question 2 is queried for the mail servers for Gmail.

Ans.

If standard DNS record is a phone book for general location, MX record is for mail room (company analogy). We want to find the location of where the mail is sent to the company location registered under @gmail.com

```
pin@MacBook-khxng-Nathan ~ % nslookup -type=MX gmail.com dns2.ox.ac.uk
Server:          dns2.ox.ac.uk
Address:         163.1.2.190#53

** server can't find gmail.com: REFUSED
```

the refusal is due to university's blockade to refuse to give out such information of their gmail server to outsider, instead with MU website.

```
[pin@MacBook-khxng-Nathan ~ % nslookup -type=MX gmail.com ldnsly.mahidol.ac.th
Server:          ldnsly.mahidol.ac.th
Address:         10.41.41.1#53

Non-authoritative answer:
gmail.com        mail exchanger = 5 gmail-smtp-in.l.google.com.
gmail.com        mail exchanger = 10 alt1.gmail-smtp-in.l.google.com.
gmail.com        mail exchanger = 30 alt3.gmail-smtp-in.l.google.com.
gmail.com        mail exchanger = 40 alt4.gmail-smtp-in.l.google.com.
gmail.com        mail exchanger = 20 alt2.gmail-smtp-in.l.google.com.

Authoritative answers can be found from:

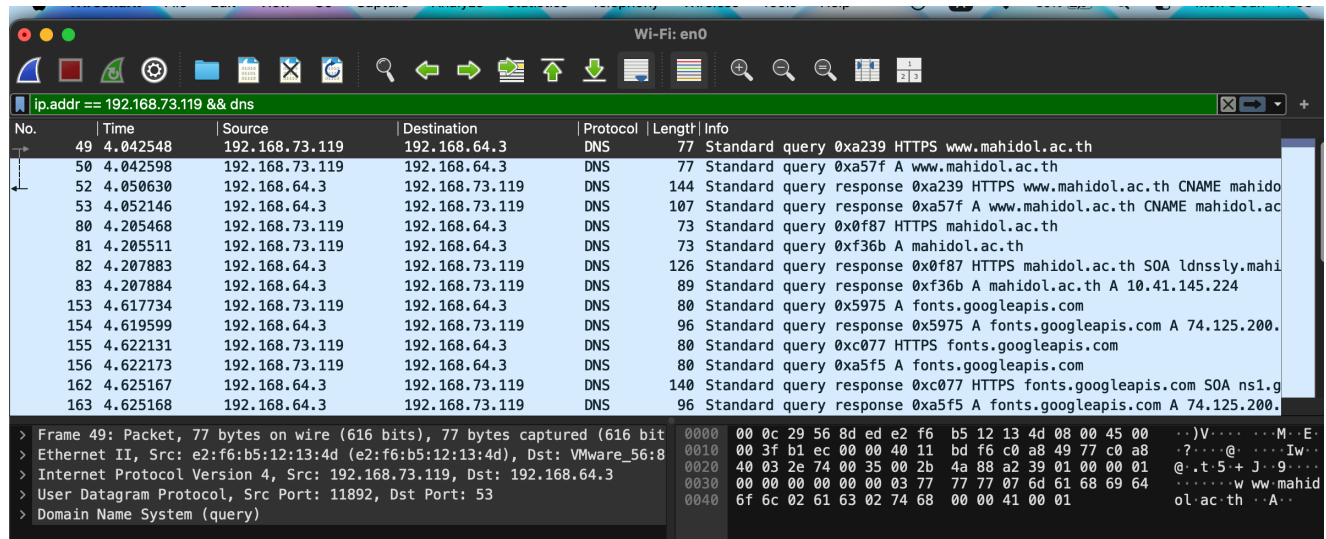
pin@MacBook-khxng-Nathan ~ %
```

Second part

- ifconfig
- sudo dscacheutil -flushcache; sudo killall -HUP mDNSResponder | for macOS to clear local DNS cache
- ifconfig en0 (this limits ifconfig to only see the information of IPs and everything, but related to local router only)

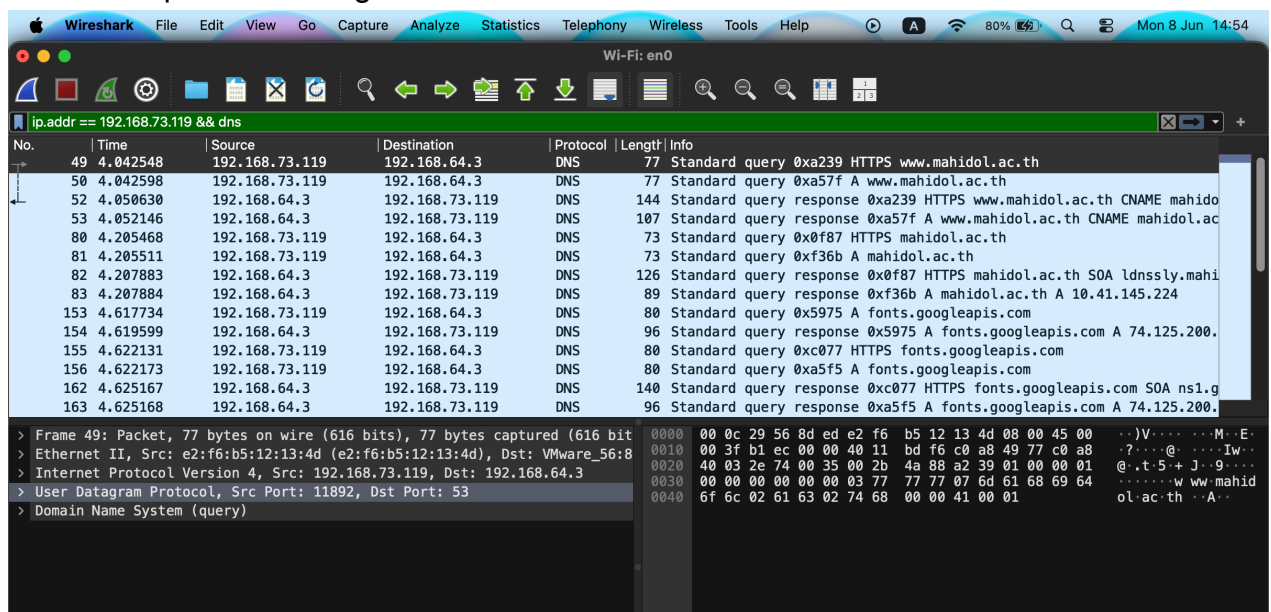
Locate the DNS query and response messages. Are they sent over UDP or TCP?

Ans.



UDP

5. What is the destination port for the DNS query message? What is the source port of DNS response message?



query dst port 53

No.	Time	Source	Destination	Protocol	Length	Info
49	4.042548	192.168.73.119	192.168.64.3	DNS	77	Standard query 0xa239 HTTPS www.mahidol.ac.th
50	4.042598	192.168.73.119	192.168.64.3	DNS	77	Standard query 0xa57f A www.mahidol.ac.th
52	4.050630	192.168.64.3	192.168.73.119	DNS	144	Standard query response 0xa239 HTTPS www.mahidol.ac.th CNAME mahidol.ac.th
53	4.052146	192.168.64.3	192.168.73.119	DNS	107	Standard query response 0xa57f A www.mahidol.ac.th CNAME mahidol.ac.th
80	4.205468	192.168.73.119	192.168.64.3	DNS	73	Standard query 0xf87 HTTPS mahidol.ac.th
81	4.205511	192.168.73.119	192.168.64.3	DNS	73	Standard query 0xf36b A mahidol.ac.th
82	4.207883	192.168.64.3	192.168.73.119	DNS	126	Standard query response 0xf87 HTTPS mahidol.ac.th SOA ldnsly.mahidol.ac.th
83	4.207884	192.168.64.3	192.168.73.119	DNS	89	Standard query response 0xf36b A mahidol.ac.th A 10.41.145.224
153	4.617734	192.168.73.119	192.168.64.3	DNS	80	Standard query 0x5975 A fonts.googleapis.com
154	4.619599	192.168.64.3	192.168.73.119	DNS	96	Standard query response 0x5975 A fonts.googleapis.com A 74.125.200.100
155	4.622131	192.168.73.119	192.168.64.3	DNS	80	Standard query 0xc077 HTTPS fonts.googleapis.com
156	4.622173	192.168.73.119	192.168.64.3	DNS	80	Standard query 0xa5f5 A fonts.googleapis.com
162	4.625167	192.168.64.3	192.168.73.119	DNS	140	Standard query response 0xc077 HTTPS fonts.googleapis.com SOA ns1.google.com
163	4.625168	192.168.64.3	192.168.73.119	DNS	96	Standard query response 0xa5f5 A fonts.googleapis.com A 74.125.200.100

> Frame 52: Packet, 144 bytes on wire (1152 bits), 144 bytes captured (1152 bits) on interface Wi-Fi: en0
 > Ethernet II, Src: VMware_56:8d:ed (00:0c:29:56:8d:ed), Dst: e2:f6:b5:12:11:3d
 > Internet Protocol Version 4, Src: 192.168.64.3, Dst: 192.168.73.119
 > User Datagram Protocol, Src Port: 53, Dst Port: 11892
 > Domain Name System (response)

response src port : 53

Ans.

6. To what IP address is the DNS query message sent? Use *ipconfig* to determine the IP address of your local DNS server. Are these two IP addresses the same?

Ans.

Query message is sent to 192.168.64.3

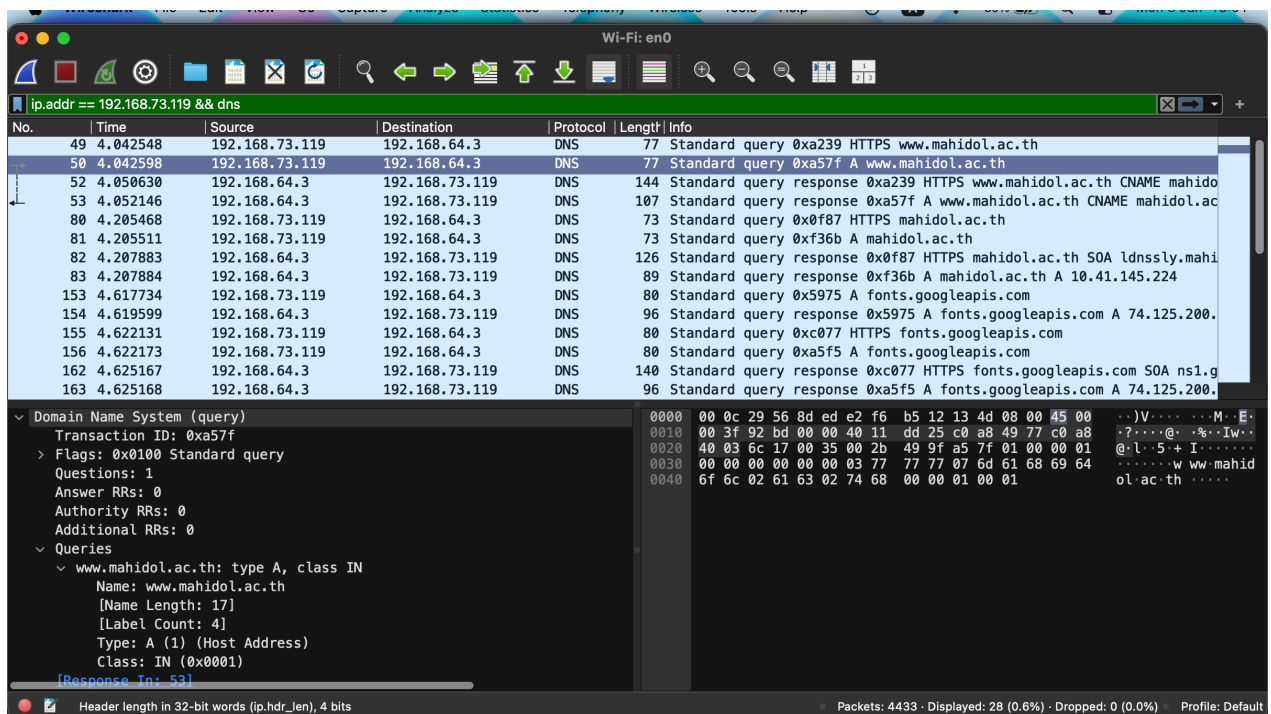
```
[pin@MacBook-khxng-Nathan ~ % cat /etc/resolv.conf
#
# macOS Notice
#
# This file is not consulted for DNS hostname resolution, address
# resolution, or the DNS query routing mechanism used by most
# processes on this system.
#
# To view the DNS configuration used by this system, use:
#   scutil --dns
#
# SEE ALSO
#   dns-sd(1), scutil(8)
#
# This file is automatically generated.
#
search muic.network
nameserver 192.168.64.3
pin@MacBook-khxng-Nathan ~ %
```

This is the IP of local DNS server

Both matches

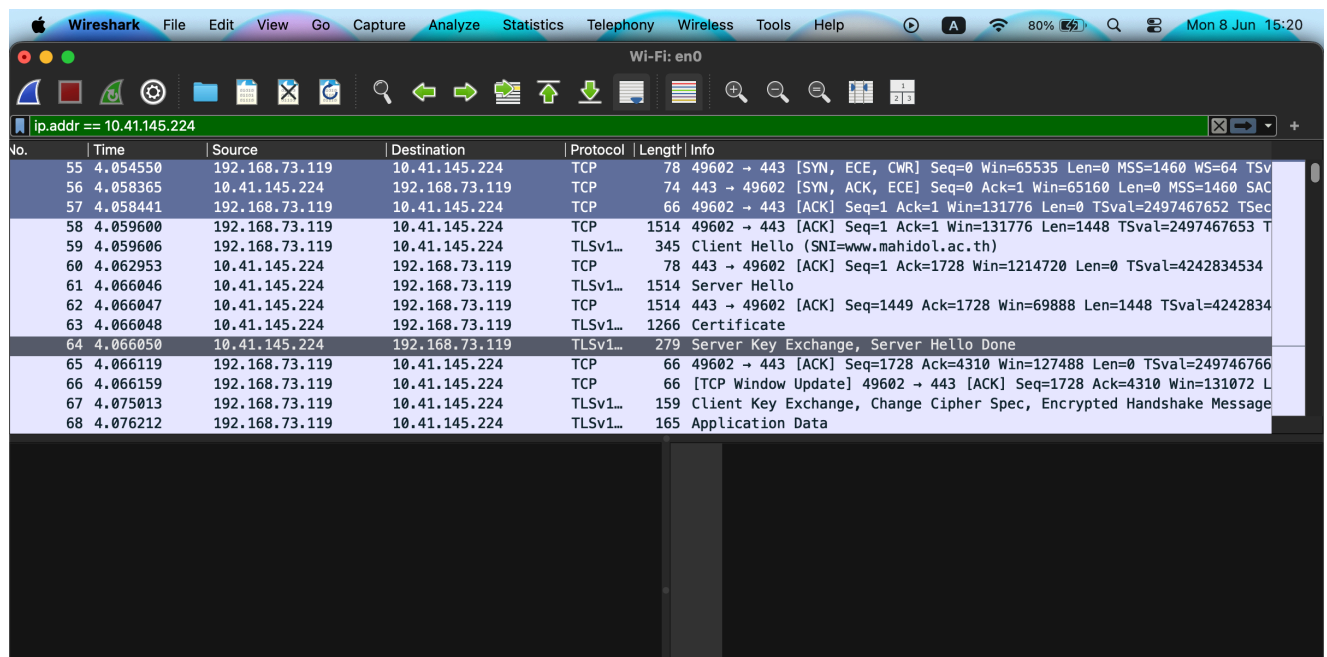
7. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

Ans.



DNS query type : Type A (address record), no answer for query

Part 3



a) What is an Initial Sequence Number (also show a captured screenshot)?

Ans.

Wi-Fi: en0

ip.addr == 10.41.145.224

No.	Time	Source	Destination	Protocol	Length	Info
55	4.054550	192.168.73.119	10.41.145.224	TCP	78	49602 → 443 [SYN, ECE, CWR] Seq=0 Win=6
56	4.058365	10.41.145.224	192.168.73.119	TCP	74	443 → 49602 [SYN, ACK, ECE] Seq=0 Ack=1
57	4.058441	192.168.73.119	10.41.145.224	TCP	66	49602 → 443 [ACK] Seq=1 Ack=1 Win=13177
58	4.059600	192.168.73.119	10.41.145.224	TCP	1514	49602 → 443 [ACK] Seq=1 Ack=1 Win=13177
59	4.059606	192.168.73.119	10.41.145.224	TLSv1...	345	Client Hello (SNI=www.mahidol.ac.th)
60	4.062953	10.41.145.224	192.168.73.119	TCP	78	443 → 49602 [ACK] Seq=1 Ack=1728 Win=12
61	4.066046	10.41.145.224	192.168.73.119	TLSv1...	1514	Server Hello
62	4.066047	10.41.145.224	192.168.73.119	TCP	1514	443 → 49602 [ACK] Seq=1449 Ack=1728 Win=12
63	4.066048	10.41.145.224	192.168.73.119	TLSv1...	1266	Certificate
64	4.066050	10.41.145.224	192.168.73.119	TLSv1...	279	Server Key Exchange, Server Hello Done
65	4.066119	192.168.73.119	10.41.145.224	TCP	66	49602 → 443 [ACK] Seq=1728 Ack=4310 Win=12
66	4.066159	192.168.73.119	10.41.145.224	TCP	66	[TCP Window Update] 49602 → 443 [ACK] S
67	4.075013	192.168.73.119	10.41.145.224	TLSv1...	159	Client Key Exchange, Change Cipher Spec
68	4.076212	192.168.73.119	10.41.145.224	TLSv1...	165	Application Data

Transmission Control Protocol, Src Port: 49602, Dst Port: 443

Source Port: 49602
Destination Port: 443
[Stream index: 3]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 4087543545
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1011 = Header Length: 44 bytes (11)
Flags: 0x0c2 (SYN, ECE, CWR)
Window: 65535

0000 c4 ad 34 e0 2d 14 e2 f6 b5 12 13 4d 08 00 45 00
0010 00 40 00 00 40 00 40 06 94 8f c0 a8 49 77 0a 29
0020 91 e0 c1 c2 01 bb f3 a2 f6 f9 00 00 00 00 b0 c2
0030 ff ff f8 1b 00 00 02 04 05 b4 01 03 03 06 01 01
0040 08 0a 94 dc 55 00 00 00 00 00 04 02 00 00

Sequence number = 0

Sequence number (raw) : 4087543545

b) What is a Sequence and Ack Number (also show a captured screenshot)?

Wi-Fi: en0

ip.addr == 10.41.145.224

No.	Time	Source	Destination	Protocol	Length	Info
55	4.054550	192.168.73.119	10.41.145.224	TCP	78	49602 → 443 [SYN, ECE, CWR] Seq=0 Win=6
56	4.058365	10.41.145.224	192.168.73.119	TCP	74	443 → 49602 [SYN, ACK, ECE] Seq=0 Ack=1
57	4.058441	192.168.73.119	10.41.145.224	TCP	66	49602 → 443 [ACK] Seq=1 Ack=1 Win=13177
58	4.059600	192.168.73.119	10.41.145.224	TCP	1514	49602 → 443 [ACK] Seq=1 Ack=1 Win=13177
59	4.059606	192.168.73.119	10.41.145.224	TLSv1...	345	Client Hello (SNI=www.mahidol.ac.th)
60	4.062953	10.41.145.224	192.168.73.119	TCP	78	443 → 49602 [ACK] Seq=1 Ack=1728 Win=12
61	4.066046	10.41.145.224	192.168.73.119	TLSv1...	1514	Server Hello
62	4.066047	10.41.145.224	192.168.73.119	TCP	1514	443 → 49602 [ACK] Seq=1449 Ack=1728 Win=12
63	4.066048	10.41.145.224	192.168.73.119	TLSv1...	1266	Certificate
64	4.066050	10.41.145.224	192.168.73.119	TLSv1...	279	Server Key Exchange, Server Hello Done
65	4.066119	192.168.73.119	10.41.145.224	TCP	66	49602 → 443 [ACK] Seq=1728 Ack=4310 Win=12
66	4.066159	192.168.73.119	10.41.145.224	TCP	66	[TCP Window Update] 49602 → 443 [ACK] S
67	4.075013	192.168.73.119	10.41.145.224	TLSv1...	159	Client Key Exchange, Change Cipher Spec
68	4.076212	192.168.73.119	10.41.145.224	TLSv1...	165	Application Data

Internet Protocol Version 4, Src: 10.41.145.224, Dst: 192.168.73.119

Transmission Control Protocol, Src Port: 443, Dst Port: 49602

Source Port: 443
Destination Port: 49602
[Stream index: 3]
[Stream Packet Number: 2]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 1600630246
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 4087543546
1010 = Header Length: 40 bytes (10)
Flags: 0x052 (SYN, ACK, ECE)
Window: 65535

0000 e2 f6 b5 12 13 4d c4 ad 34 e0 2d 14 08 00 45 00
0010 00 3c 00 00 40 00 3c 06 98 93 0a 29 91 e0 c0 a8
0020 49 77 01 bb c1 c2 5f 67 ad e6 f3 a2 f6 fa a0 52
0030 fe 88 7c 71 00 00 02 04 05 b4 04 02 08 0a fc e4
0040 84 61 94 dc 55 00 01 03 03 07

Ans. Sequence number = 0, Acknowledgement number = 1 (relative ack number)

c) What is an Ack Number (also show a captured screenshot)?

The screenshot shows a Wireshark packet capture of a TLS handshake. The packet list pane displays the following packets:

No.	Time	Source	Destination	Protocol	Length	Info
55	4.054550	192.168.73.119	10.41.145.224	TCP	78	49602 → 443 [SYN, ECE, CWR] Seq=0 Win=6
56	4.058365	10.41.145.224	192.168.73.119	TCP	74	443 → 49602 [SYN, ACK, ECE] Seq=0 Ack=1
57	4.058441	192.168.73.119	10.41.145.224	TCP	66	49602 → 443 [ACK] Seq=1 Ack=1 Win=13177
58	4.059600	192.168.73.119	10.41.145.224	TCP	1514	49602 → 443 [ACK] Seq=1 Ack=1 Win=13177
59	4.059606	192.168.73.119	10.41.145.224	TLSv1...	345	Client Hello (SNI=www.mahidol.ac.th)
60	4.062953	10.41.145.224	192.168.73.119	TCP	78	443 → 49602 [ACK] Seq=1 Ack=1728 Win=12
61	4.066046	10.41.145.224	192.168.73.119	TLSv1...	1514	Server Hello
62	4.066047	10.41.145.224	192.168.73.119	TCP	1514	443 → 49602 [ACK] Seq=1449 Ack=1728 Win=12
63	4.066048	10.41.145.224	192.168.73.119	TLSv1...	1266	Certificate
64	4.066050	10.41.145.224	192.168.73.119	TLSv1...	279	Server Key Exchange, Server Hello Done
65	4.066119	192.168.73.119	10.41.145.224	TCP	66	49602 → 443 [ACK] Seq=1728 Ack=4310 Win=12
66	4.066159	192.168.73.119	10.41.145.224	TCP	66	[TCP Window Update] 49602 → 443 [ACK] S
67	4.075013	192.168.73.119	10.41.145.224	TLSv1...	159	Client Key Exchange, Change Cipher Spec
68	4.076212	192.168.73.119	10.41.145.224	TLSv1...	165	Application Data

The packet details pane for packet 57 shows the following information:

- Internet Protocol Version 4, Src: 192.168.73.119, Dst: 10.41.145.224
- Transmission Control Protocol, Src Port: 49602, Dst Port: 443
 - Source Port: 49602
 - Destination Port: 443
 - [Stream index: 3]
 - [Stream Packet Number: 3]
 - > [Conversation completeness: Incomplete, DATA (15)]
 - [TCP Segment Len: 0]
 - Sequence Number: 1 (relative sequence number)
 - Sequence Number (raw): 4087543546
 - [Next Sequence Number: 1 (relative sequence number)]
 - Acknowledgment Number: 1 (relative ack number)
 - Acknowledgment number (raw): 1600630247
 - 1000 = Header Length: 32 bytes (8)
 - > Flags: 0x010 (ACK)

The packet bytes pane shows the raw data of the packet, with the first 32 bytes highlighted in blue.

Ans.

Acknowledgement number = 1 (and raw number is as followed in the screenshot)